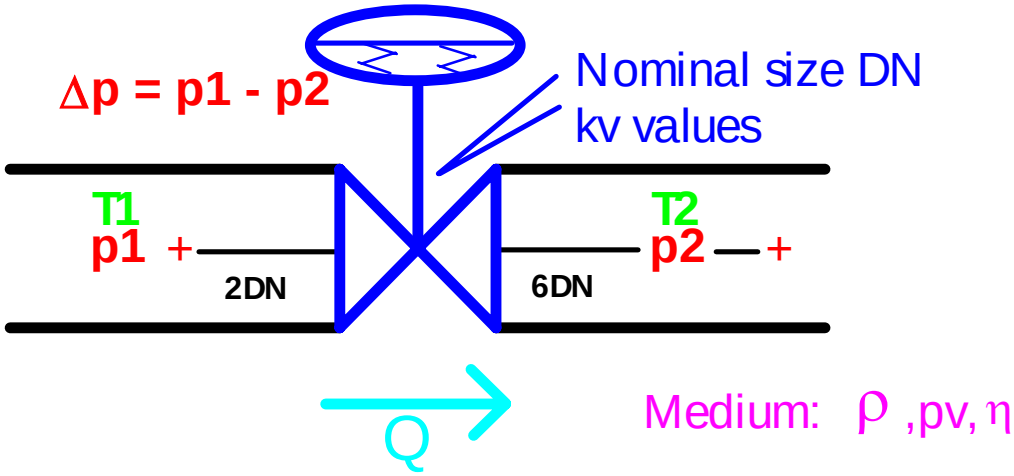


3. Sizing flow capacity

3. Sizing flow capacity

3.1. Relation between pressure loss and the kv value definition



Experience shows that the pressure difference Δp with a throttle element and a turbulent flow is proportional to the quadratic flow quantity (Q^2).

In flow technics, one usually uses the so-called pressure-loss coefficient ζ for this purpose, which is always assigned to a cross-section A (e.g. nominal valve size cross section):

$$\Delta p = p_1 - p_2 = \frac{\rho}{2} \bullet \zeta \bullet \left(\frac{Q}{A_{\text{nom. size}}}\right)^2$$

In automated engineering, process quantities are controlled by changing the flow quantity Q. The pressure difference is simply a means to this end (valve authority). As a parameter for flow capacity, one therefore has the k_v value as the water quantity k_v in m^3/h at a pressure difference of $\Delta p_0 = 1\ \text{bar}$. The water density at 20 degrees C is $\rho_0 = 1000\ kg/m^3$.

$$\Delta p_0 = p_1 - p_2 = \frac{\rho_0}{2} \bullet \zeta \bullet \left(\frac{k_v}{A_{\text{nom. size}}}\right)^2$$

or

$$\zeta = \frac{2 \bullet \Delta p_0}{\rho_0} \bullet \left(\frac{A_{\text{nom. size}}}{k_v}\right)^2$$

The last equation gives the relation between the pressure loss coefficient ζ (with relation to the nominal size) and the k_v value.

The rule of thumb usually applied:

$$k_v = \frac{Q}{\sqrt{\Delta p}}$$

is only correct for water (20 degrees C). More correct is the formulation:

$$k_v = \sqrt{\frac{\Delta p_0}{\Delta p}} \bullet \frac{\sqrt{\rho}}{\sqrt{\rho_0}} \bullet Q$$

or

$$Q = \sqrt{\frac{\Delta p}{\Delta p_0}} \bullet \frac{\sqrt{\rho_0}}{\sqrt{\rho}} \bullet k_v \quad .$$

This equation means that the flow quantity doubles when the pressure difference is increased four times.

The equation above is only correct for non-compressible media such as water. Gaseous and vaporous media are compressible, so one must account for density changes through the flow path using a correction factor, the so-called expansion factor Y. If one uses the inlet density ρ_1 and the flow volume Q_1 at the valve entrance, one arrives at the following equation:

$$k_v = \sqrt{\frac{\Delta p_0}{\Delta p}} \bullet \frac{\sqrt{\rho_1}}{\sqrt{\rho_0}} \bullet Q_1 \bullet \frac{1}{Y}$$

Due to mass conservation during passage of the valve, the inlet flow mass is equal to the outlet flow mass. Due to the pressure-dependent density, the flow volume on the inlet side (Q_1) is less than on the outlet side (Q_2). It is a good idea to use the flow mass $\dot{m} = W = W_1 = W_2$.

$$k_v = \sqrt{\frac{\Delta p_0}{\Delta p}} \bullet \frac{\sqrt{\rho_1}}{\sqrt{\rho_0}} \bullet \frac{W}{\rho_1} \bullet \frac{1}{Y}$$

The expansion factor is less than 1. Therefore, greater k_v values are required than for liquids with the same operating and materials data.

Due to additional limiting conditions (cavitation, speed of sound), this correction factor is not the only one. The equations required are contained in Parts 2-1 and 2-2 of the DIN IEC 534 standard. Due to the non-perspicuous form used there, the unit-independent form has always been selected here, and one basic equation is used for liquids and gases/vapors.

3.2.DIN IEC 534 P. 2-1, 2-2 and 2-3

These parts are important for the sizing of a control valve with respect to flow capacity.

Part 2-1(2-2): Determination of flow capacity (k_v value) or flow Q (W)
Part 2-3: Test procedure for experimental determination of the k_v value.

The basic equation mentioned earlier is:

$$k_v = \sqrt{\frac{\Delta p_0}{\Delta p}} \bullet \frac{\sqrt{\rho_1}}{\sqrt{\rho_0}} \bullet Q_1 \bullet \frac{1}{F_P \bullet F_R \bullet Y}, \quad \Delta p \le \Delta p_{max}$$

with

$$\rho_0 = 1000\ kg/m^3 \quad \text{and} \quad \Delta p_0 = 1\ bar$$

The correction factors F_P, F_R and Y take into account the following influences:

Flow limitation: Δp_{max}, velocity throttling point

The influence of pipeline geometry: F_P

Expansion factor: Y

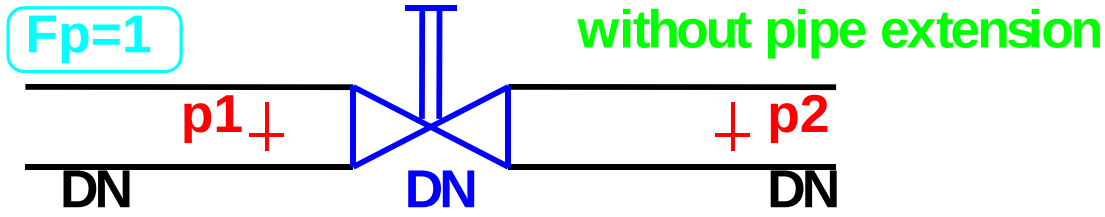
Viscosity influence: F_R

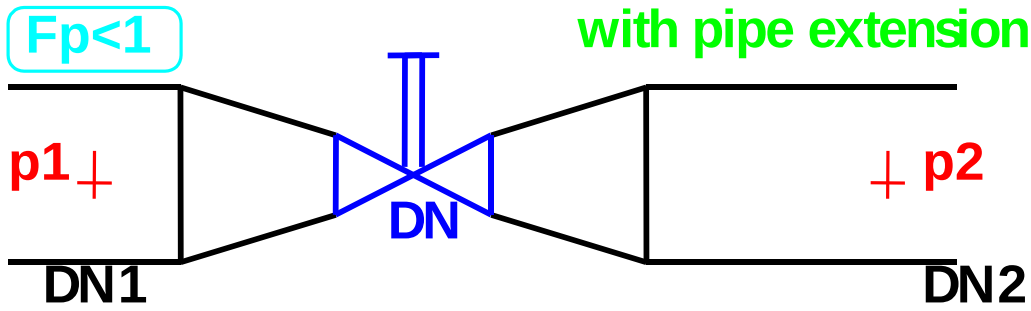
3.2.1.Pipeline geometry factor F_P

The k_v valve value relates to a continuous, straight pipeline in front of and behind the valve. The pressure reduction points relate to minimum distances of 2 nominal valve sizes in front of and 6 nominal valve sizes behind the valve, in order to minimize the inflow and outflow effects of the flow. However, if the valve is connected to the rest of the pipeline system with fittings, it must be seen as a unit by the system planner, i.e. the k_v then refers to the valve with fittings. The valve manufacturer, however, is less interested in the k_v value with pipe extension than in the k_v value of the valve. This is why the pipeline geometry factor F_P is introduced. It represents the relation between these two k_v values. It can be estimated by applying the energy equation for the individual fittings. More exact values can only be obtained by measurements (DIN IEC 534 P. 2-3).

The F_P value is less than 1 and decreases above all for valves with higher specific flow outlet (k_v / D_N^2), i.e. for butterfly valves and ball valves. Linear control valves can usually be calculated well with $F_P = 1$.

$$F_P = \frac{k_{v,\text{ with pipe extension}}}{k_v} \le 1$$





Especially for approximate calculations:

$F_P \cong 1$ if D_N / D_1 or $D_N / D_2 > 0.8$ *and* $\frac{k_v}{D_N^2} < 0.02$ $k_v (m^3/h)$ and D_N (mm)

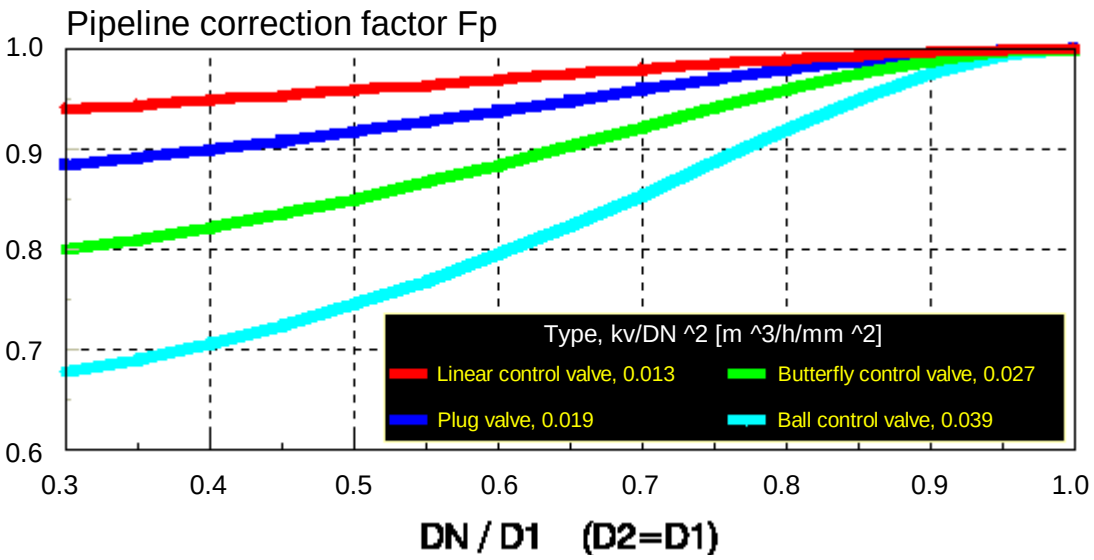
Generally

$$F_P = \sqrt{1 - \frac{\rho_0}{2\Delta p_0} \cdot (\zeta_{B1} + \zeta_1 - \zeta_{B2} + \zeta_2) \cdot \left(\frac{k_{v,\text{ with pipe extension}}}{\frac{\pi}{4} \cdot D_N^2}\right)^2}$$ otherwise

Pressure loss or energy conversion coefficients

$$\zeta_{B1} \approx 1 - \left(\frac{D_N}{D_{N1}}\right)^4 \qquad \zeta_1 \approx \frac{1}{2} \cdot \left(\left(1 - \left(\frac{D_N}{D_{N1}}\right)^2\right)^2\right)$$

$$\zeta_{B2} \approx 1 - \left(\frac{D_N}{D_{N2}}\right)^4 \qquad \zeta_2 \approx \left(\left(1 - \left(\frac{D_N}{D_{N2}}\right)^2\right)^2\right)$$



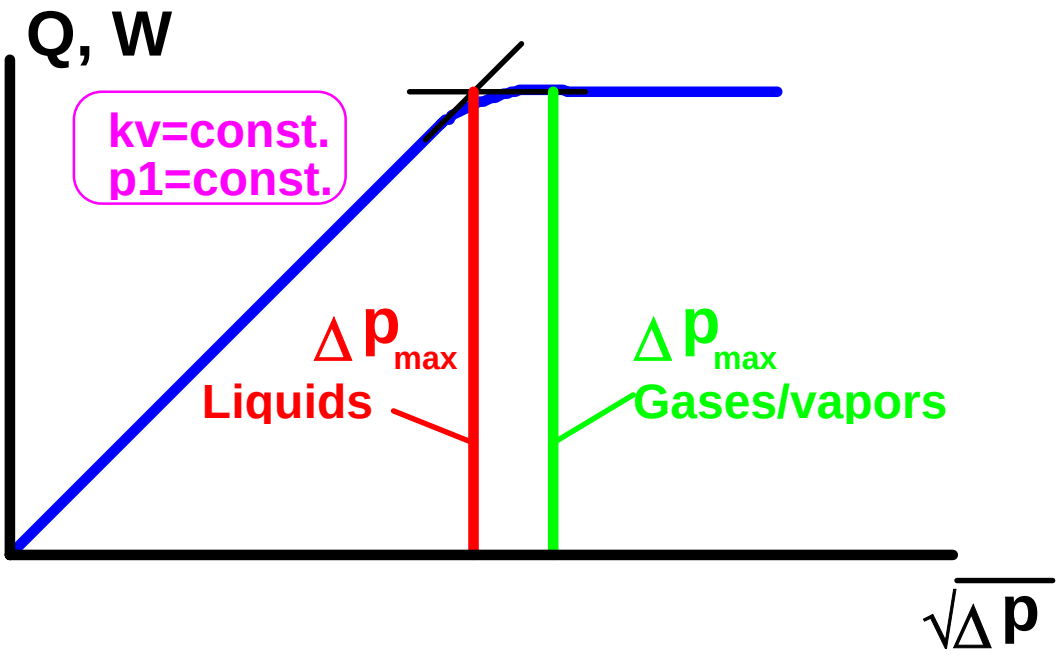
3.2.2. Flow limitation (Δp_{max})

Flow limitation (i.e. no increase of flow quantity at constant inlet pressure p_1 and constant k_v value despite increasing pressure difference) arises when

the speed of sound is reached with gas or vapor flows in the throttling point

and when heavy cavitation ($p_2 \rightarrow p_v$) or flashing is reached with liquid flows.

The corresponding critical pressure difference Δp_{max} is calculated according to medium type.



Non-compressible media (liquids)

$$\Delta p_{max} = \frac{F_{LP}^2}{F_P^2} \bullet (p_1 - F_F \bullet p_v)$$

$$F_{Lp} = F_L \qquad \text{,if } F_P = 1$$

$$F_{Lp} = \frac{F_L}{\sqrt{1 + \frac{\rho_0}{2\Delta p_0} \cdot F_L^2 \cdot (\zeta_{B1} + \zeta_1) \cdot (\frac{k_v}{\frac{\pi}{4} \cdot D_N^2})^2}} \qquad \text{,if } F_P < 1$$

$$F_F = 0.96 - 0.28 \bullet \sqrt{\frac{p_v}{p_c}} \qquad p_c\text{: critical pressure}$$

The F_L value is a valve parameter. It is referred to as the pressure recovery factor. Linear control valves have the highest F_L values (at 0.9 to 0.95) and therefore larger, more useful critical pressure differences for flow limitation than other valve types.
This value must be corrected (F_{LP}) if fittings are present and $F_P < 1$ is therefore the case.

Compressible media (gases, vapors)

$$\Delta p_{max} = x_{TP} \bullet \frac{\gamma}{1.4} \bullet p_1$$

$$x_{TP} = x_T \qquad \text{,if } F_p = 1$$

$$x_{TP} = \frac{\frac{x_T}{F_P^2}}{1 + \frac{\rho_0}{2\Delta p_0} \cdot \frac{8}{9} \cdot x_T \cdot (\zeta_{B1} + \zeta_1) \cdot (\frac{k_v}{\frac{\pi}{4} \cdot D_N^2})^2} \qquad \text{, if } F_P < 1$$

The x_T value is a valve parameter. It is designated as the critical pressure ratio for flow mass limitation. Linear control valves have the highest x_T values (at 0.68 to 0.77) and therefore larger, more useful critical pressure differences for flow limitation than other valve types.
This value must be corrected (x_{TP}) if fittings are present and $F_P < 1$ is therefore the case.

x_T value is strongly correlated to the F_L value ($x_T \cong 0.86 \bullet F_L^2$).

The simultaneous ones of the F_L value for pressure recovery and flow limitation can be explained in the following manner:

The pressure loss Δp is proportional to the velocity energy in the vena contracta (throttling point) $\rho / 2 \bullet u_{vc}^2$ (as with Carnot thrust loss)

$$\Delta p = \zeta_{Carnot} \bullet \frac{\rho}{2} \bullet u_{vc}^2 \quad \text{with } \zeta_{Carnot} = F_L^2$$

The velocity energy is obtained approximately using the Bernoulli equation and ignoring pressure losses from inlet 1 to the throttling point vc.

$$p_1 - p_{vc} \cong \rho / 2 \bullet u_{vc}^2 \quad \text{(Bernoulli)}$$

The following relation results

$$\frac{\Delta p}{p_1 - p_{vc}} = F_L^2$$

When flow limitation has just been reached, the pressure in the trottling point is equal to the critical pressure $p_{vc,crit} = F_F \bullet p_v$, and the pressure difference is Δp_{max}.

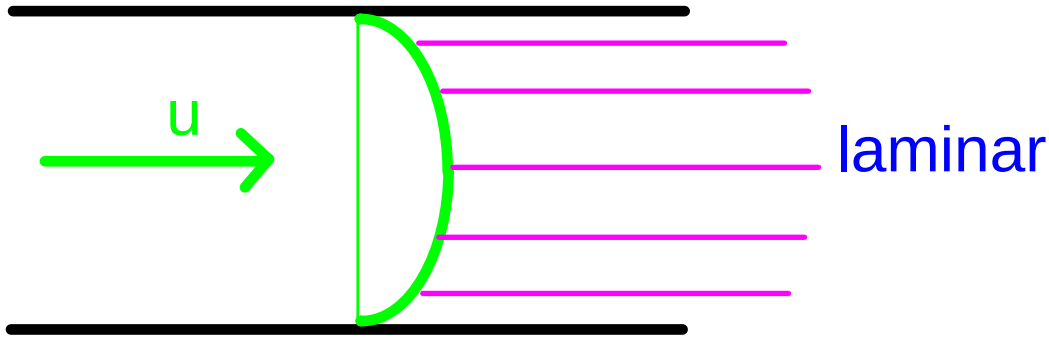
$$\frac{\Delta p_{max}}{p_1 - p_{vc,crit}} = F_L^2$$

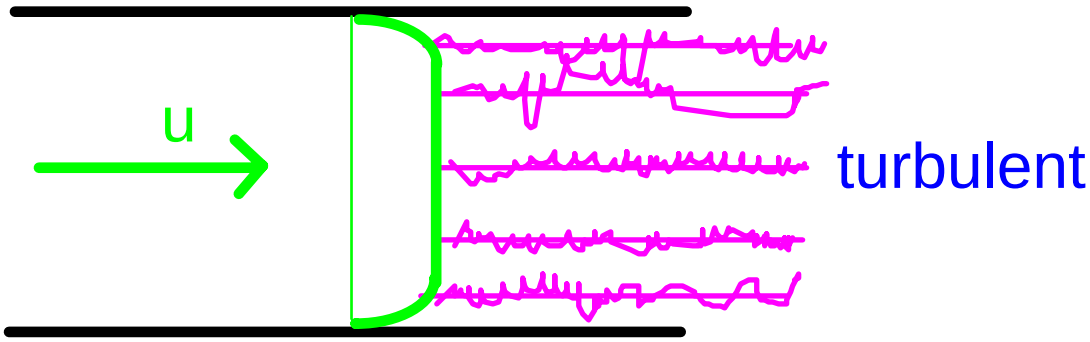
A higher pressure recovery means that at a fixed velocity u_{vc} in the throttling point and a fixed inlet pressure p_1, the pressure difference Δp is small or the pressure p_2 is great. This means the same as with a small F_L value, but also the same as with achieving flow limitation at lower pressure differences (disadvantage with butterfly valves).

3.2.3. Influence of viscosity (correction factor $F_R(Re$, Reynolds number Re)

In flow technics, one differentiates in principle between laminar and turbulent flow conditions, with almost 100% of all valve applications running turbulently.

Laminar flows arise in some circumstances with very viscous (thick) flow media, very small valve dimensions (microvalves) or with very small flow quantities. The are characterized by an ordered flow almost without chaotic motions lateral to the direction of flow.





The so-called valve Reynolds' number is a judgement measure for whether a flow is turbulent.

This dimensionless parameter combines

the geometry dimensions (throttle diameter dependent on the k_v value, the F_L value and the valve form factor F_d)
the kinematic viscosity
and the flow quantity Q

Such Reynolds' numbers are used in flow technics for pipe and split flows, for example.

Valve Reynolds' number:

$$Re = \frac{2^{5/4}}{\pi^{1/2} \cdot \upsilon} \bullet \left(\frac{\Delta p_0}{\rho_0}\right)^{1/4} \bullet \frac{Q \cdot F_d}{\sqrt{k_v \cdot F_L}} = 1{,}34 \bullet \left(\frac{\Delta p_0}{\rho_0}\right)^{1/4} \bullet \frac{Q \cdot F_d}{\nu \bullet \sqrt{k_v \cdot F_L}}$$

The valve form factor F_d accounts for the geometric form of the throttling point in the form of the hydraulic diameter d_{hyd} as the diameter d_0 (throttle cross-section area converted into circle surface area). The hydraulic cross-section is defined as the quadruple throttle cross-section area divided by the circumference of the jet emitted by the throttling point. It characterizes the ratio of the jet surface area (when one also considers the jet length) to the flow cross-section. The total resistance force resulting from the transverse stresses in effect in the flow (viscosity), and therefore the pressure loss, is dependent upon this.

Example: Pipeline (diameter d_0):

$$d_h = \frac{4 \bullet \frac{\pi}{4} \bullet d_0^2}{\pi \bullet d_0 \bullet L} = d_0 \qquad F_d = \frac{d_h}{d_0} = 1$$

Example: Annular gap (gap width s, diameter S_b, $s << S_b$, microvalve):

$$d_h = \frac{4 \bullet \pi \bullet s \bullet S_b}{2 \bullet \pi \bullet S_b} = 2s \qquad F_d = \sqrt{\frac{s}{S_b}}$$

For valve Reynolds' numbers greater than 10,000, experience shows that turbulent flow conditions are always present. The correction factor F_R here is always 1.

Below 10,000 there is an interim range to lower F_R values, before laminar flow conditions set in. Because the pressure loss for laminar flows is $\Delta p \sim Q$ or Re, the correction factor is $F_R \sim \sqrt{Re}$.

In contrast to older versions of DIN IEC 534 P. 2-1, the correction factor procedures for the constant K depend on the specific flow outlet $k_v / D_N^{\,2}$ (see below). Numerous measurements were carried out especially for SAMSON microvalves to allow the most exact sizing possible. These were also included in DIN IEC 534 and were applied there to all valves types generally with a certain amount of uncertainty. In this program, the SAMSON Type 3510 Microvalve was calculated with an approximation curve for

$$K = f(k_v / D_N^{\,2})$$

which approximates the measurements.

Equations for FR

turbulent range $F_R(Re) = 1$, for $Re \geq 10000$

interim range

$$F_R(Re) = \left(\frac{Re}{10000}\right)^{\frac{1}{1+\log(Re)+K}}$$

for $Re < 10000$,
minimum laminar range

$$F_R(Re) = 0.026 \bullet \sqrt{K \bullet Re} / F_L$$

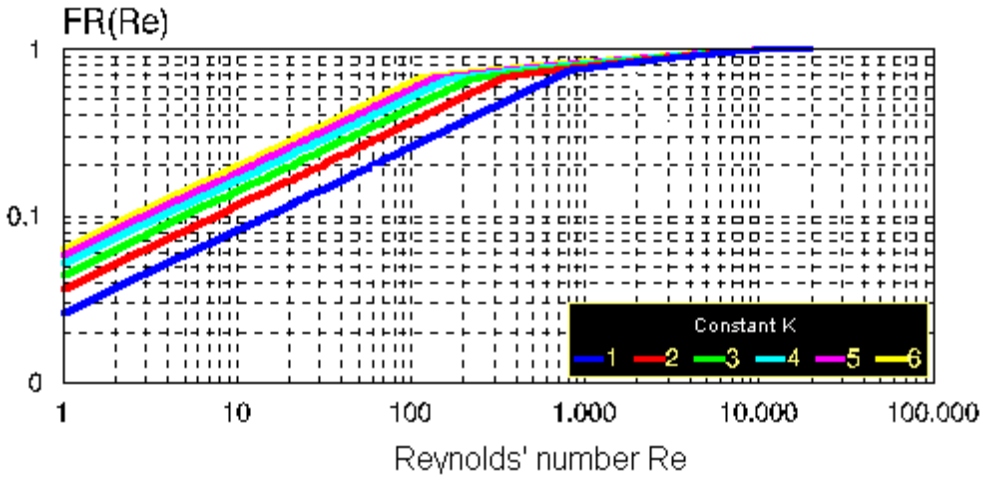
if $F_R \neq 1$: Pipeline geometry factor $F_P = 1$

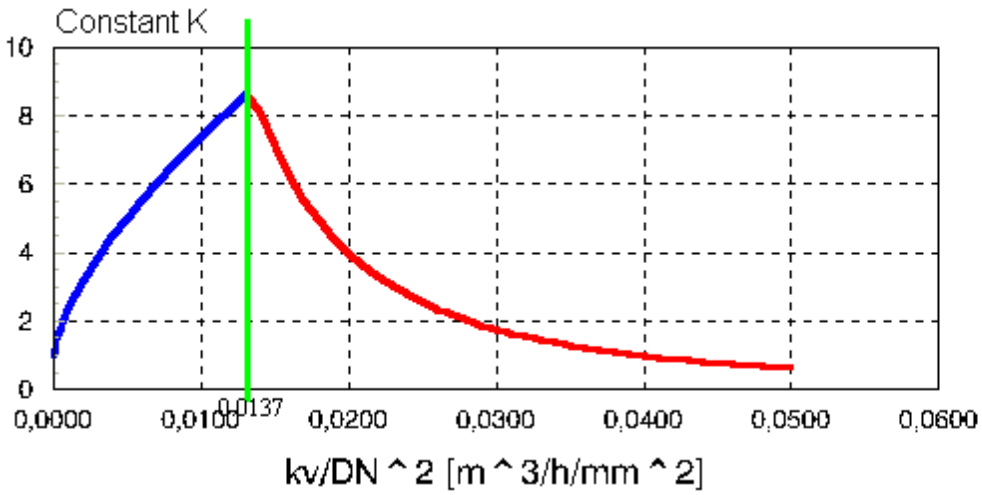
Constant K

$$\frac{kv}{DN^2} \leq 0.0137 \quad (kv\ [m^3/h],\ D_N\ [mm]) \qquad K = 1 + 138 \bullet \left(\frac{kv}{DN^2}\right)^{\frac{2}{3}}$$

$$\frac{kv}{DN^2} > 0.0137 \quad (kv\ [m^3/h],\ D_N\ [mm]) \qquad K = 0.0016 \bullet \frac{1}{\left(\frac{kv}{DN^2}\right)^2}$$

Special: SAMSON Type 3510 Microvalve specially adapted to the measurements





One sees that greater corrections are necessary for smaller specific flow outlets ($K->1$) than for higher flow outlets.
The correction factor can only be determined iteratively.

3.2.4. Expansion factor Y

Non-compressible media (liquids)

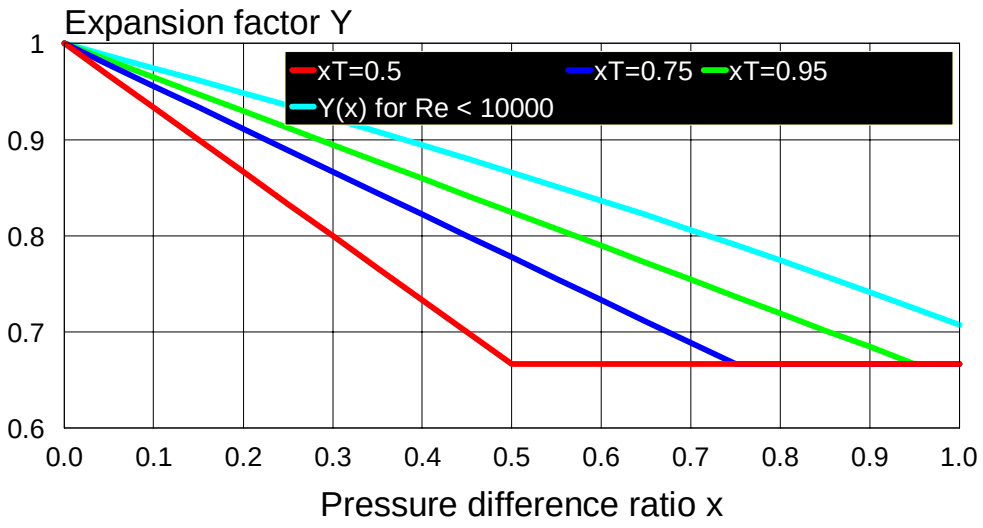
$$Y(x) = 1$$

Compressible media (gases, vapors)

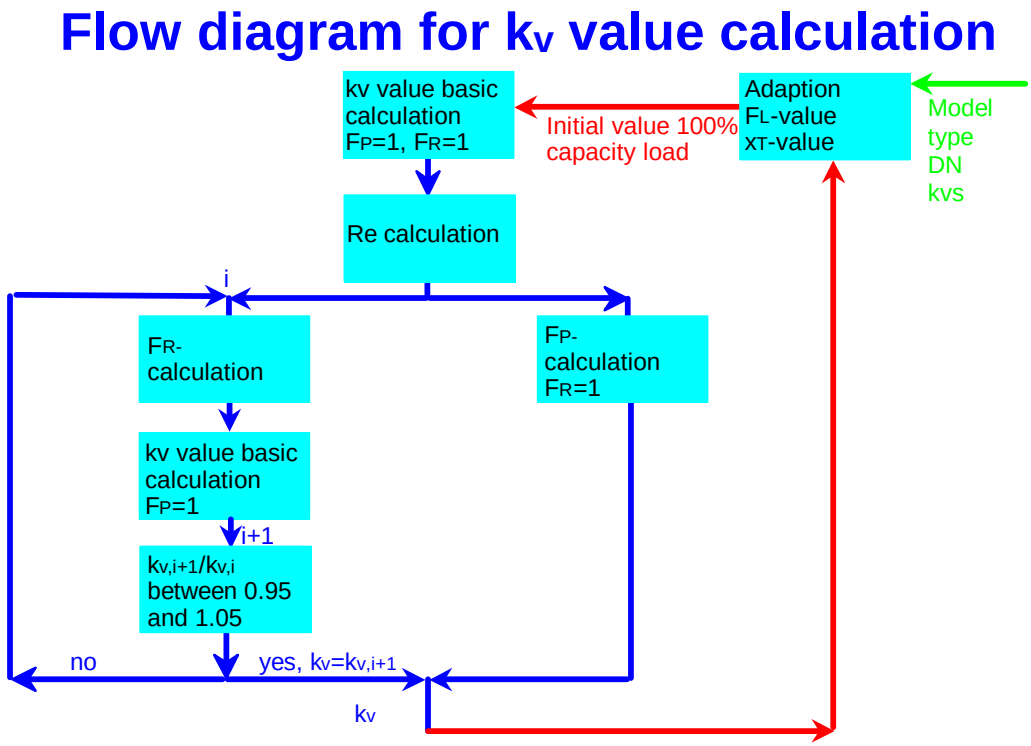
$$Y(x) = 1 - \frac{x}{3 \bullet xT} \geq \frac{2}{3}, \qquad Re \geq 10000$$

$$Y(x) = \sqrt{1 - \frac{1}{2} \bullet x}, \qquad Re < 10000$$

Pressure difference ratio $x = (p_1 - p_2) / p_1$



3.2.5. Sequence of kv value determination



3.2.6. Flow characteristics

Usually, three operating points for the minimum, standard and maximum quantity ranges must be considered for the valve outlet. This assumes knowledge of the operating and materials data. Based upon this, 3 k_v values are calculated k_{vmin}, k_{vnorm} and k_{vmax} for a given valve type. The k_{vs} value is suggested in the valve sizing program together with a safety factor (1.1 (10%) is standard).

$$k_{vs} = k_{v\,max} \bullet S_F$$

Valve parameters such as F_L, x_T are dependent on valve type, k_{vs} value and the nominal size D_N, so that iterations which the user does not notice occur during calculation in the background in the valve sizing program.

To fulfill the prescribed control task, the condition

$$k_{v\,min} > k_{vs} / Ra \quad , \quad Ra: \text{Rangeability value}$$

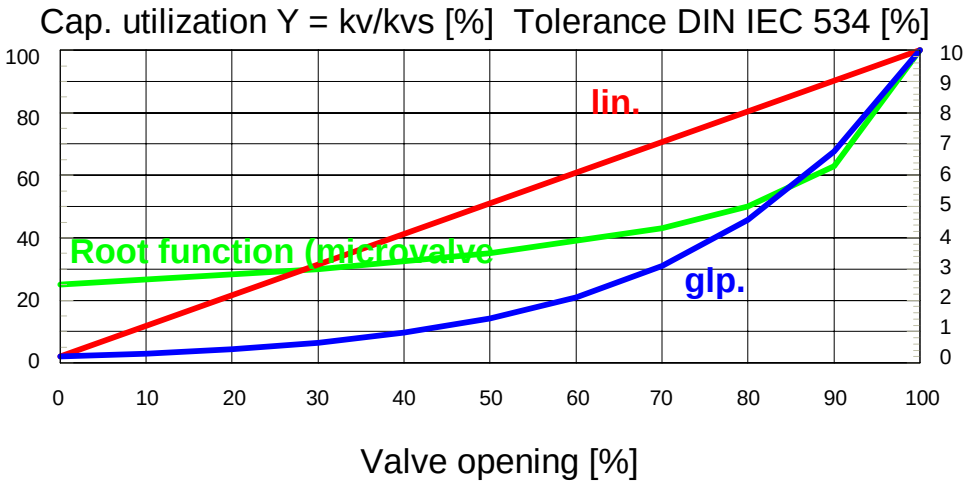
must also be fulfilled while also observing a characteristic form (e.g. linear, of equal percentage).

Typical rangeabilities can be taken from the following table.

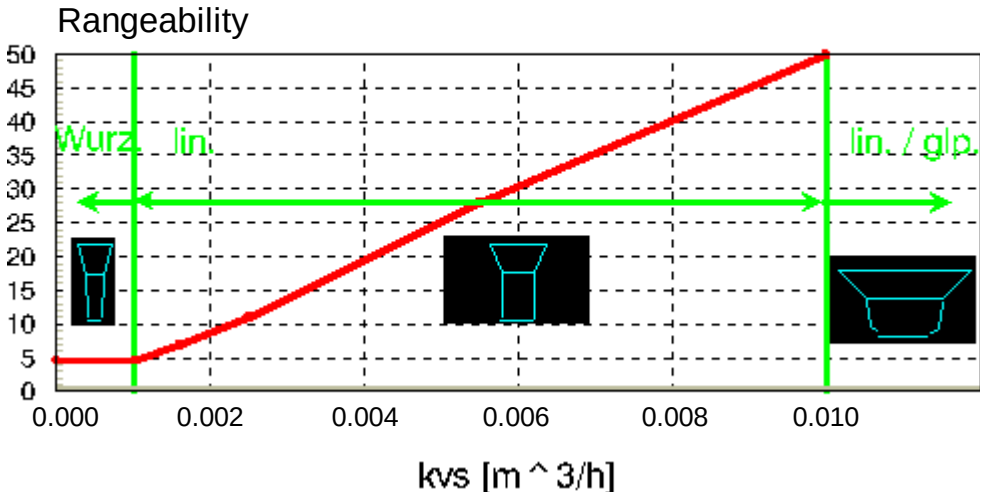
Control valve type	**Linear**	**Same percentage**	**Root funktion**
Linear control valve	30:1 to 50:1 (contour)	0:1 to 50:1 (contour)	
Microvalve	5:1 to 50:1 (contour)	50:1 (contour)	5:1
Butterfly valve	50:1 (cam disk)	50:1 (cam disk)	
Plug valve	150-200:1	150-200:1 (cam disk)	

For linear control valves with $k_{vs} > 0.01\,m^3/h$, linear and same percentage characteristic forms can be implemented by adapting the ball contour.
At k_{vs} values $< 0.001\,m^3/h$ micro control valves have almost cylindrical annular gap forms in the throttling area, so that the rangeability must necessarily decrease with the usual rated travel distances. The flow characteristic then degenerates into a so-called root function characteristic. In the lowest k_{vs} value range, the flow usually changes to a laminar condition, so that the rangeability is usually squared (from $k_v \bullet F_R \sim \sqrt{Re}$ and $Q \sim Re$, it follows that $Q \sim k_{v^2}$).

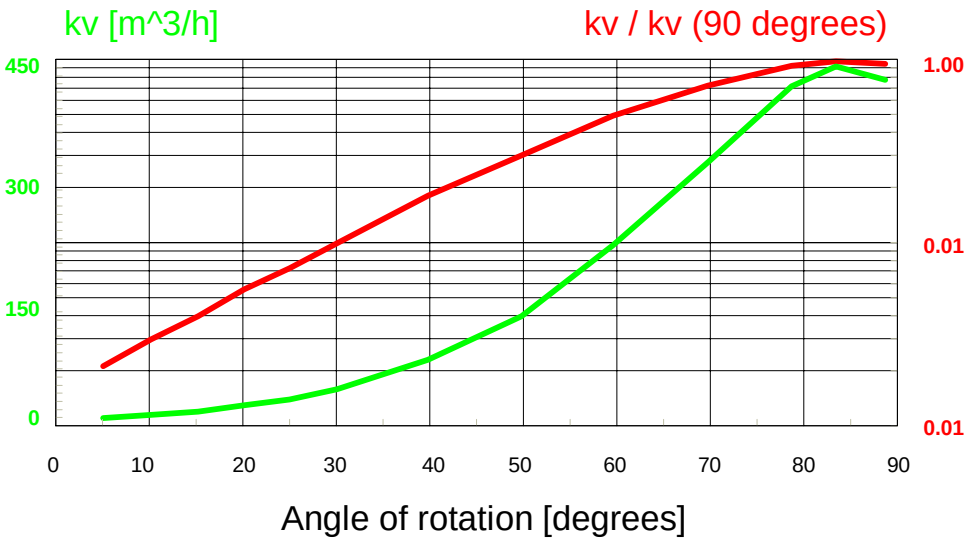
Without a cam disk in the positioner, butterfly valves have a tendency to be same percentage. Plug valves tend toward linear flow characteristics, with the commonly propagated rangeabilities of > 50:1 being heavily exaggerated because the characteristic tolerance according to DIN IEC 534 P. 2-4 cannot be fulfilled.



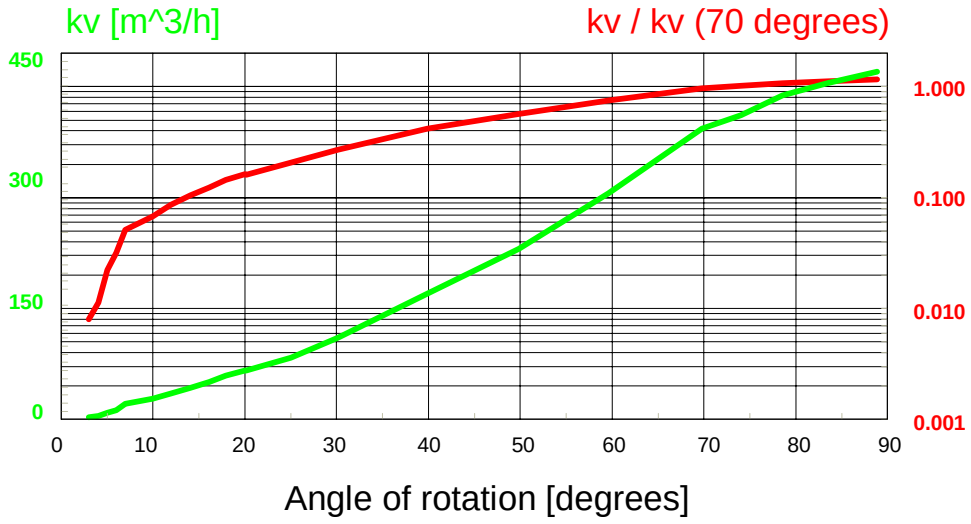
Rangeabilities (microvalve)



k_v characteristics for butterfly valves



k_v characteristic for plug valves



3.2.7. Determining the correct nominal valve size

The correct nominal valve size D_N is obtained from a maximum authorized limit velocity u_{2limit} in the valve outlet cross-section. These limit values are based on values gained by experience, but they can also be changed by the user in the valve sizing program.

The average velocity for a selected nominal valve size is (W: mass flow):

$$u_2 = \frac{W}{\frac{\pi}{4} \bullet D_N^2 \bullet \rho_2}$$

The required nominal valve size D_{Nerf} can be determined by converting the equation above:

$$D_{Nerf} = \sqrt{\frac{W}{\frac{\pi}{4} \bullet u_{2limit} \bullet \rho_2}}$$

Limit values for liquids without flashing ($p_2 > p_v$):

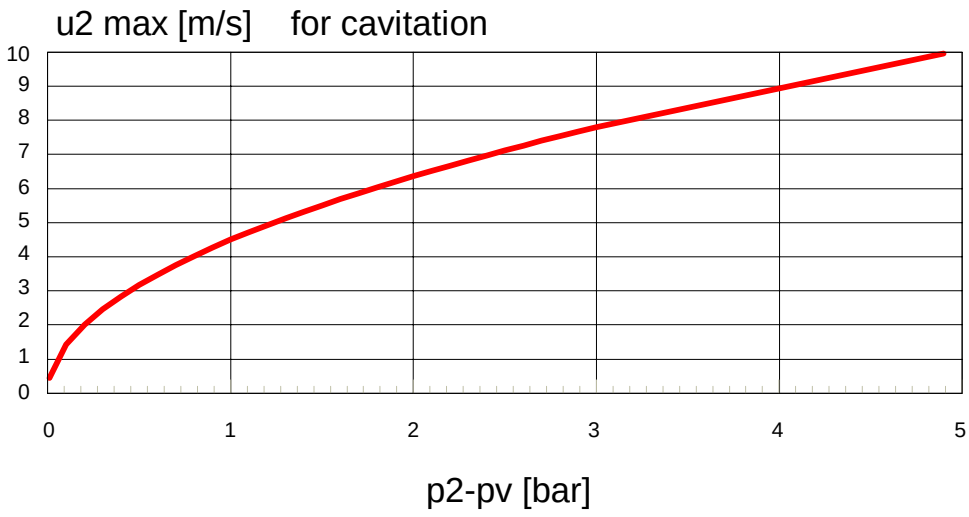
Density: $\rho_2 = \rho_1$

always: $u_{2limit} = 10\ m/s$

but also system-dependent, e.g. power plant area, heating equipment... $u_{2limit} = 1\ m/s$

additionally in cavitation area

$u_{2limit} = 4.5 \bullet \sqrt{[(p_2 - p_v) / 1bar]} \bullet 1m/s$, p_v: Vapor pressure



Limit values for gases and vapors

ρ_2: Gas equation or vapor table (see program documentation, materials data)
Mach number limitation in valve outlet

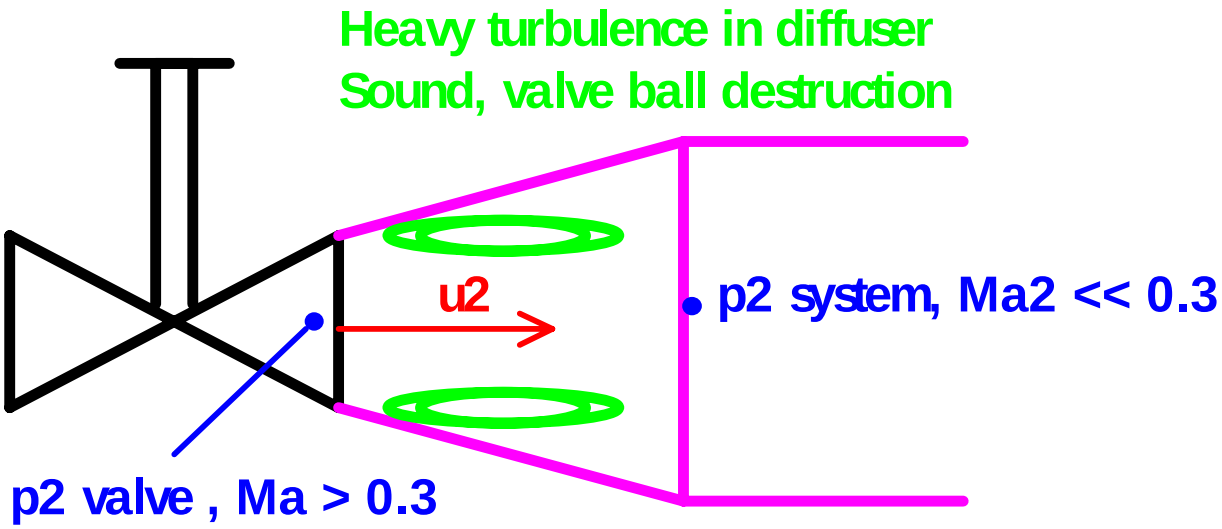
$$Ma_{2limit} = u_2 / c_2 < 0.3$$

$$\text{or } u_{2limit} = 0.3 \cdot c_2$$

Speed of sound c_2 in valve outlet:

$$c_2 = \sqrt{\gamma \cdot p_2 / \rho_2}$$

with diffuser-type pipe extensions behind the valve, the p_2 pressure used internally in the program (p_{2valve}) is less than the p_2 pressure in the large pipeline.



Liquids with flashing ($p_2 \leq p_v$, $p_1 > p_v$):

always: $u_{2limit} = 60\ m/s$

Proportion evaporated x_{d2} in % of weight for ***water*** from the energy equation

$$x_{d2} = \frac{(\frac{p_1 - p_v}{\rho_1} + h_1' - h_2')}{h_2'' - h_2'} \bullet 100\ \%$$

Inlet 1: $p_v = f(T)$, outlet 2: $p_v = p2$
': Liquid proportion, '': Wet vapor proportion,

Enthalpies h: Approximate equations (from vapor proportion)

Specific volume v_2'' (wet vapor) for ***water*** from approximate equation (vapor chart)

For media other than water: Enter x_{d2} and v_2'' directly

Outlet density

$$\rho_2 = \frac{1}{(1 - \frac{x_{d2}}{100\%}) \bullet \frac{1}{\rho_1} + \frac{x_{d2}}{100\%} \bullet v_2''}$$

Liquids with flashing and vapor proportion xd1 at inlet ($p_2 \leq p_v,\ p_1 \approx p_v$)

as above $u_{limit} = 60\ m/s$

Proportion evaporated x_{d2} in % of weight for ***water*** from the energy equation

$$x_{d2} = \frac{(\frac{p_1 - p_v}{\rho_1} + h_1' - h_2' + (h_1'' - h_1') \cdot \frac{x_{d1}}{100\%})}{h_2'' - h_2'} \bullet 100\ \%$$

Outlet density as above

3.2.8. Calculation of the k_v value for two-phase flows

There is no standard calculation procedure for this at this time. A prerequisite for liquid/vapor mixtures is that the vapor mass proportion x_{d1} at inlet is known. This also applies to liquid/gas mixtures.

The most simple calculation procedure is the k_v addition model with a correction factor $F_{cor,2ph}$, with the two phase flows handled separately. This yields two individual k_v values ($k_{v,f}$ l: Liquid, $k_{v,d,g}$: vapor/gas), which are added.

$$k_{v,fl} = \sqrt{\frac{\Delta p_0}{\rho_0}} \bullet \sqrt{\frac{\rho_{fl}}{\Delta p}} \bullet \frac{W \bullet (1 - \frac{x_{d,g,1}}{100\%})}{\rho_{fl}} \bullet \frac{1}{(F_P \bullet F_R \bullet Y)_{,fl}}, \qquad \Delta p \le (\frac{F_{LP}}{F_P})^2 \bullet (p_1 - F_F \bullet p_v)$$

$$k_{v,d,g} = \sqrt{\frac{\Delta p_0}{\rho_0}} \bullet \sqrt{\frac{\rho d,g,1}{\Delta p}} \bullet \frac{W \bullet \frac{x_{d,g,1}}{100\%}}{\rho_{d,g,1}} \bullet \frac{1}{(F_P \bullet F_R \bullet Y)_{,d,g}}, \qquad \Delta p \le x_{TP} \bullet \frac{\gamma}{1.4} \bullet p_1$$

$$k_{v,add} = (k_{v,fl} + k_{v,d,g}) \bullet F_{cor,2ph}$$

As with flashing, the pressure difference is limited purely mathematically by the critical value for flow limitation.

The added k_v value must now be multiplied by an additional safety factor $F_{cor,2ph}$, because the two phases do not flow independently of each other within the valve and a velocity compensation between the "fast" gas or vapor phase and the "slow" liquid phase takes place.
A targeted correction can be achieved with the *Sheldon and Schuder* procedure.

$$F_{cor,2ph,Sheldon\ /\ Schuder} = (1 + M_a \cdot M_p \cdot F_m(v_1'))$$

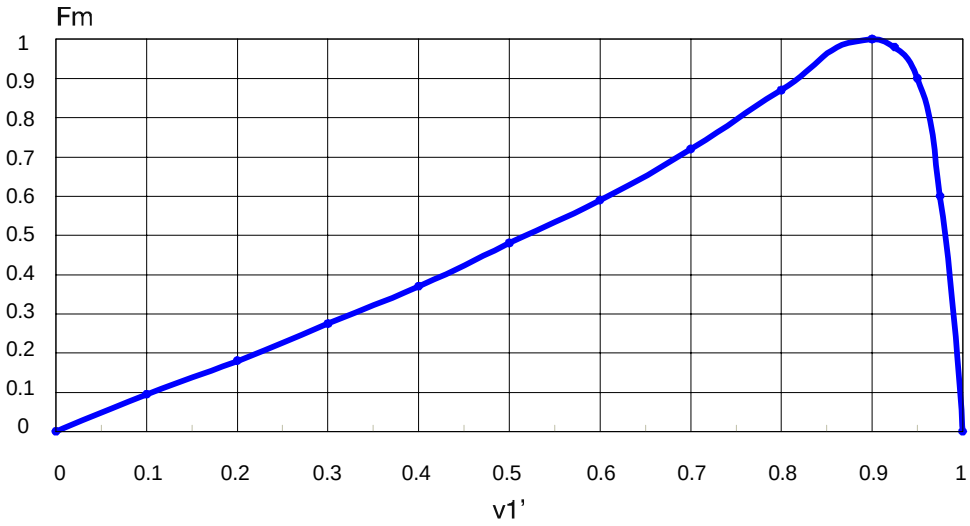
Volume content: $$v_1' = \frac{\frac{x_{d,g1}}{\rho_{d,g,1}}}{(\frac{1 - x_{d,g1}}{\rho_{fl,1}} + \frac{x_{d,g,1}}{\rho_{d,g,1}})} = \frac{V_{d,g}}{V_{d,g} + V_{fl}}$$

$$M_p = 0.35 + 0.65 \cdot \frac{A_{vc}}{A_{DN}} \quad \text{with} \quad A_{vc} = k_v \cdot F_L \cdot (\frac{\rho 0}{2 \cdot \Delta p_0})^{\frac{1}{2}}$$

$$Ma = 0.75 + 0.5 \cdot x \quad \text{for } x \le 0.5 \quad \text{with } x = (p_1 - p_2) \ / \ p_1$$

$$Ma = 1 \quad \text{for } x > 0.5$$

F_m from diagram



This function is not integrated at this time, but will definitely be contained in later program versions.

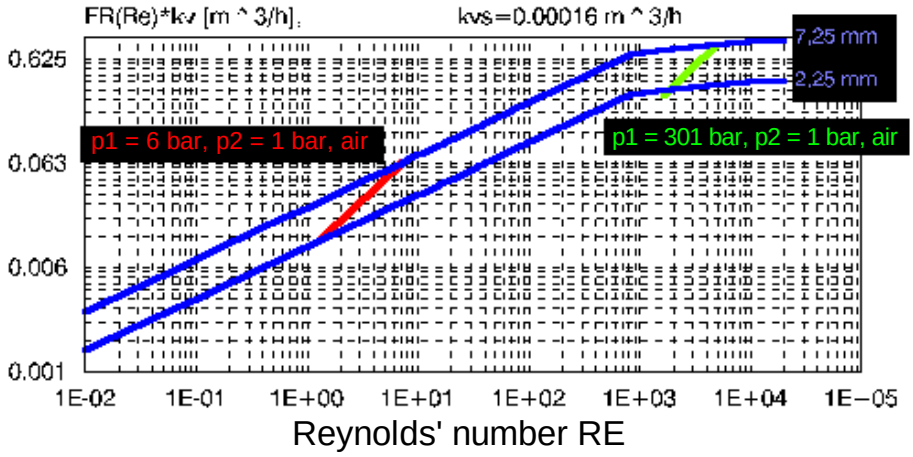
SAMSON AG, working in cooperation with flow technics department of the TU Hamburg-Harburg (L. Friedel, Dr. Engineering), has carried out detailed investigations on the valve flow behavior of water/steam systems. This has resulted in the development of a calculation procedure which is

certainly the most accurate available today, but which will also be available in a later version.

At present, the program user must create 2 files (e.g. 60dg.pos and 60fl.pos) for each measurement point for the liquid and the gaseous/vaporous parts. He then obtains 2 k_v values which must be added together. 1.35 should be used in the interim as the correction factor $F_{cor,2ph}$.

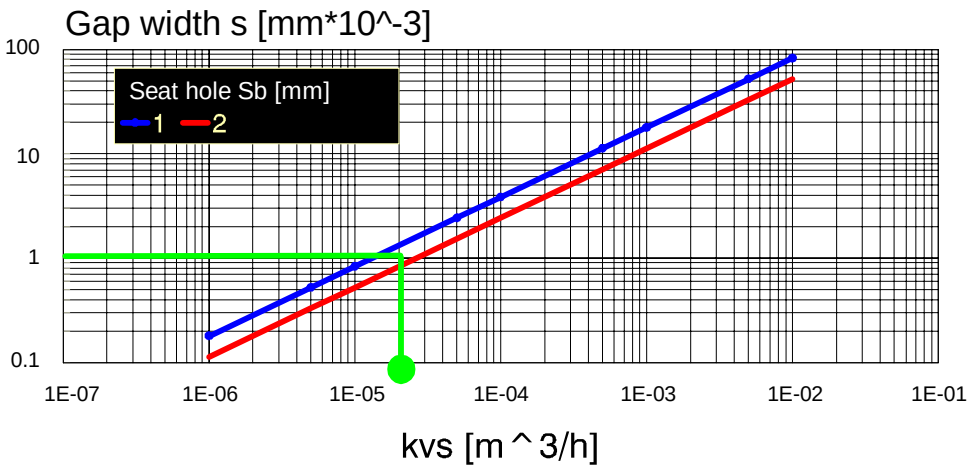
3.2.9. Lower limit of K_{vs} values

Micro control valves regularly show K_{vs} values of 10^{-6} or 10^{-7} m^3/h, which are based on air quantity measurements with pressurized air at 6 bar. However, this does not account for the fact that laminar flow conditions are present in the K_{vs} value range of 10^{-5} m^3/h. This means that the viscosity correction factor F_R is significantly smaller than 1 and that the K_{vs} value must therefore be corrected upwards.



Turbulent conditions ($F_R = 1$) do not arise here until there are significantly greater pressure differences.

Estimates with a theoretical model show that K_{vs} values below 10^{-5} m^3/h require gap widths smaller than $1\ \mu m$, even when seat holes of 1 or 2 mm are used. This cannot be practically implemented for reasons of manufacture or can only be implemented without long-term stability (wear).


Gap width s [mm*10^-3]
100
10
1
0.1
Seat hole Sb [mm]
1
2
1E-07
1E-06
1E-05
1E-04
1E-03
1E-02
1E-01
kvs [m ^ 3/h]